

# Shrish Mahesh

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## EDUCATION

### Purdue University

*Bachelor of Science in Computer Science and Mathematics*

- **GPA:** 4.0/4.0

West Lafayette, IN

Aug. 2023 – May 2027

## EXPERIENCE

### AI Software Developer Intern

June 2026 – Present

*SAP*

*San Ramon, CA*

- Designed and developed a **Splunk API observability dashboard** tracking usage trends, per-customer API breakdowns, resource-intensive endpoints, and failure modes to improve operational visibility across internal teams
- Refactored **54 Playwright UI automation test cases** and identified **4 critical UI defects**, reducing flaky failures and improving pre-release confidence
- Resolved **configuration-driven data issues** in test environments that had caused failures for months, using **AI-assisted code exploration** to navigate an unfamiliar large codebase and trace root causes to instance-level configuration changes
- Built **Bash** automation scripts to auto-resolve configuration-driven data issues, removing need for manual intervention

### Undergraduate Research Assistant

Nov 2025 – Present

*IDEAS Lab - Purdue University*

*West Lafayette, IN*

- Submitted co-author paper on Visual Interactive Manipulation for Robots to **IEEE RA-L**
- Labeled **2,000+** images contributing to a **10,000 image grasp affordance dataset**, a core contribution of the paper, for training a VLM for automated grasp labeling
- Developed preprocessing pipelines to convert and clean **point cloud datasets** and benchmarked existing grasp pose generation methods against our approach for the **IEEE RA-L** paper

### Artificial Intelligence Intern

May 2025 – Aug. 2025

*Tata Communications*

*Chennai, India*

- Engineered a retrieval evaluation framework for the company's **RAG-based** Text-to-SQL pipelines using **LlamaIndex** and open-source **LLMs**, enabling systematic evaluation with engineer-written and synthetic evaluation prompts
- Developed a **Streamlit** dashboard to track usage metrics of 4 different Text-to-SQL and Text-to-Mongo pipelines and deployed on the company's internal cloud with **Docker**; enabled quicker and more effective pipeline oversight
- Built and refactored a **Text-to-SQL pipeline API** using **FastAPI** and **Streamlit**, working across a large codebase; implemented streaming responses to eliminate blank-screen wait time and improve output digestibility

### Undergraduate Researcher - Open Source LLM Sub-team Lead

Jan. 2025 – Dec 2025

*Purdue Vertically Integrated Projects*

*West Lafayette, IN*

- Led a sub-team of 8 building a **RAG-enhanced** LLM tutor for a Data Science course using **open-source LLMs**
- Evaluated **5 open-source models** on QA performance; implemented PDF parsing and data reformatting pipelines
- Built text and image-based retrieval pipelines feeding into an **LLM/VLM**, with image-based retrieval achieving **50%** higher document relevance
- Integrated **OpenAI API** for diagram generation and multimodal inputs; built a **CoT-based planning tool** that decomposes user requests, routes to appropriate tools and agents, and prompts users for confirmation before execution

## PROJECTS

### ClearDex | *Python, Gemini Robotics, OpenAI API, ROS2, Gazebo, GPD*

Mar. 2026 – May 2026

- Built an end-to-end semantic table-clearing pipeline in **ROS 2** and **Gazebo**, integrating **Gemini Robotics-ER**, **SAM2**, and **GPT-4o-mini** to drive a simulated **Franka Panda** arm
- Built a two-stage perception pipeline using **Gemini Robotics-ER** for object localization, **SAM2** for instance segmentation; **100%** accuracy on segmentation and semantic labeling
- Generated structured **JSON** task plans with **GPT-4o-mini**; executed pick-and-place via **MoveIt 2** and **GPD** with **96%** task planning accuracy across 5 trials

### Grade Saver | *Python, Git, Playwright, Flask, Llama, Qwen, OpenCV, HTML, CSS, JavaScript*

April 2025 – Present

- Won Best Browser Automation award at Catapult (AI startup hackathon) with **200+** competing participants
- Contributed to an **AI-powered Chrome extension** that scrapes Gradescope rubrics and auto-fills regrade requests with **LLM-generated** arguments for appropriate rubric items
- Built a preprocessing and prompting pipeline using **OpenCV** contrast enhancement and **Qwen2.5 VL** for text extraction from student submissions, feeding context to **Llama3.1** for regrade argument generation

## TECHNICAL SKILLS

**Languages:** Java, Python, C, C++, SQL (MySQL), JavaScript, HTML/CSS, x86-64 Assembly

**Frameworks:** Flask, Django, FastAPI, LlamaIndex, ROS2, MoveIt 2, Gazebo

**Libraries:** Pandas, NumPy, Plotly, Scikit-learn, OpenCV, Tensorflow, Unsloth, Hugging Face, Streamlit, Gymnasium, SAM2

**Tools:** Git, GitHub, Linux, AWS, Docker, Claude Code